SmartRoute: City Transit & Delivery Dispatch Dashboard

A DATA STRUCTURES AND ALGORITHMS (DSA) PROJECT
& OPTIMIZING DISPATCH & ROUTING SYSTEMS

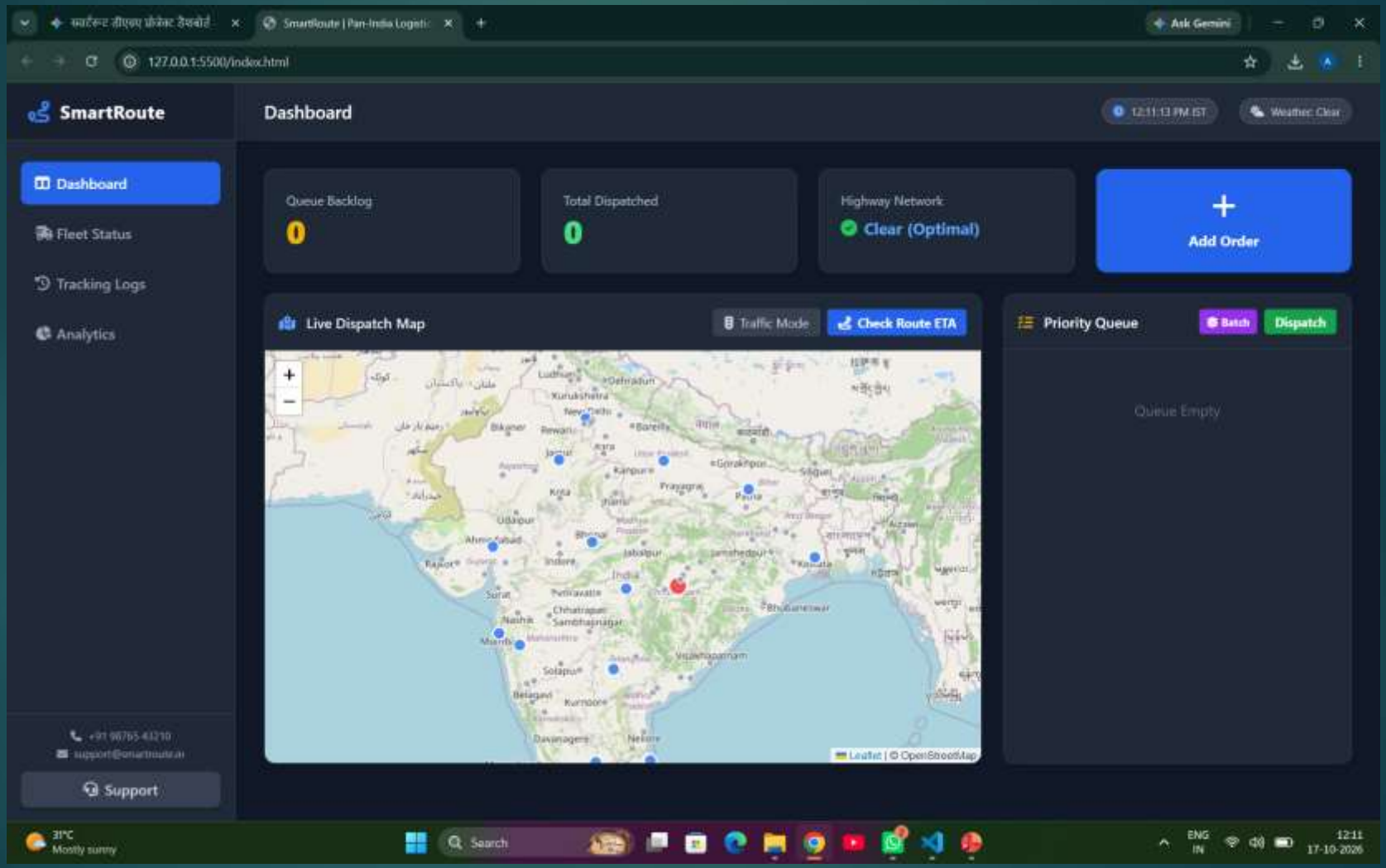
Project Overview

- **What is SmartRoute?**
  - An interactive, enterprise-grade fleet management dashboard built to visualize and manage pan-India logistics.
- **Core Objective**
  - To practically implement theoretical DSA concepts (Graphs, Min-Heaps) in solving real-world routing and dispatch prioritization problems.
- **Scale & Scope**
  - Manages deliveries across 115+ Indian cities, dynamically calculating optimal paths, estimated costs, and real-time ETAs.

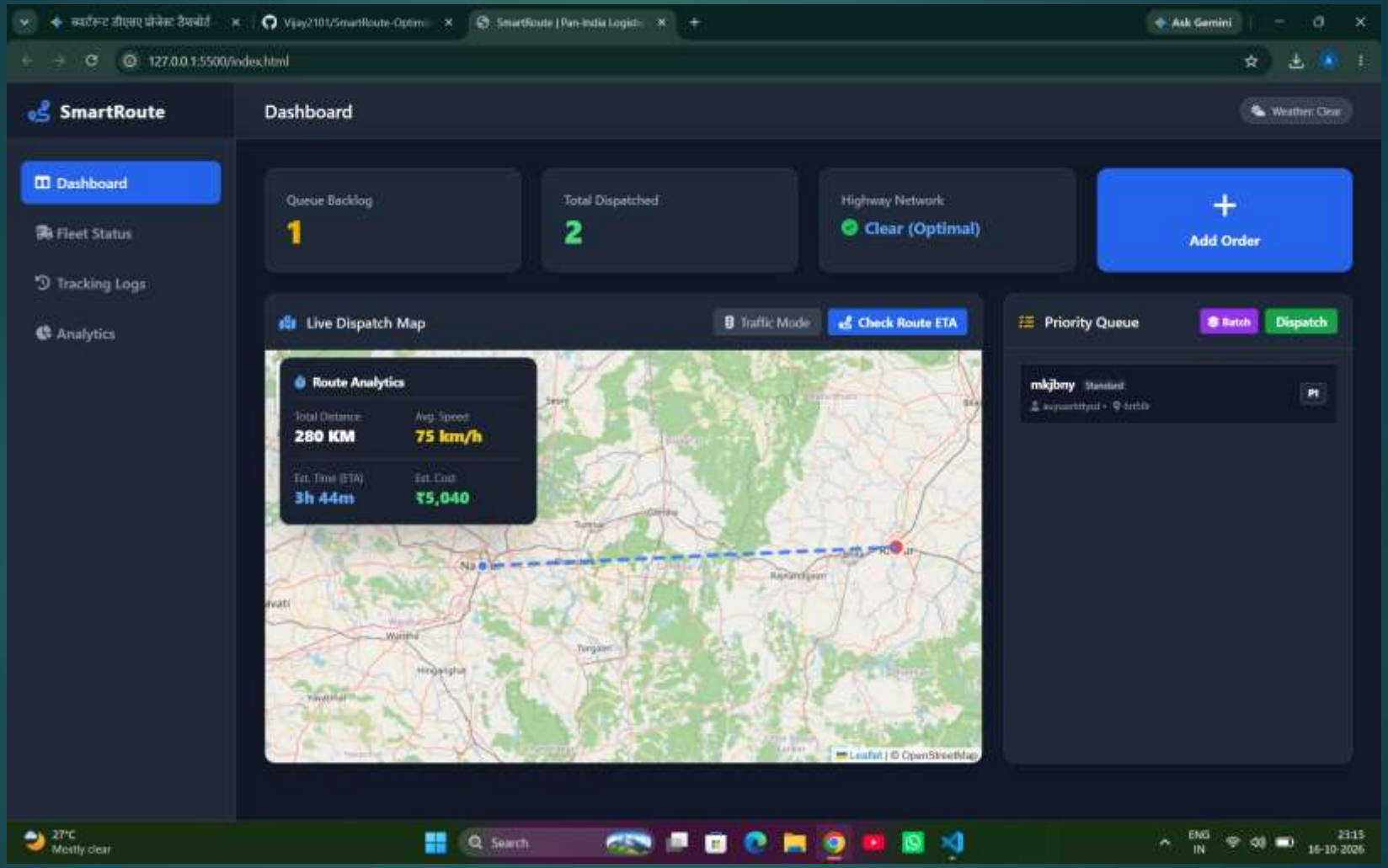
Key DSA Concepts Implemented

- **Dijkstra's Algorithm (Graph Theory)**
  - Computes the shortest and most cost-effective path from the Headquarters to any destination hub. Edge weights dynamically adapt to simulated traffic and weather conditions.
- **Priority Queue (Min-Heap Array Logic)**
  - Dynamically sorts and manages pending orders based on urgency levels (P1-Urgent, P2-Express, P3-Standard). Ensures emergency deliveries are dispatched first.

Live Dispatch Map & UI



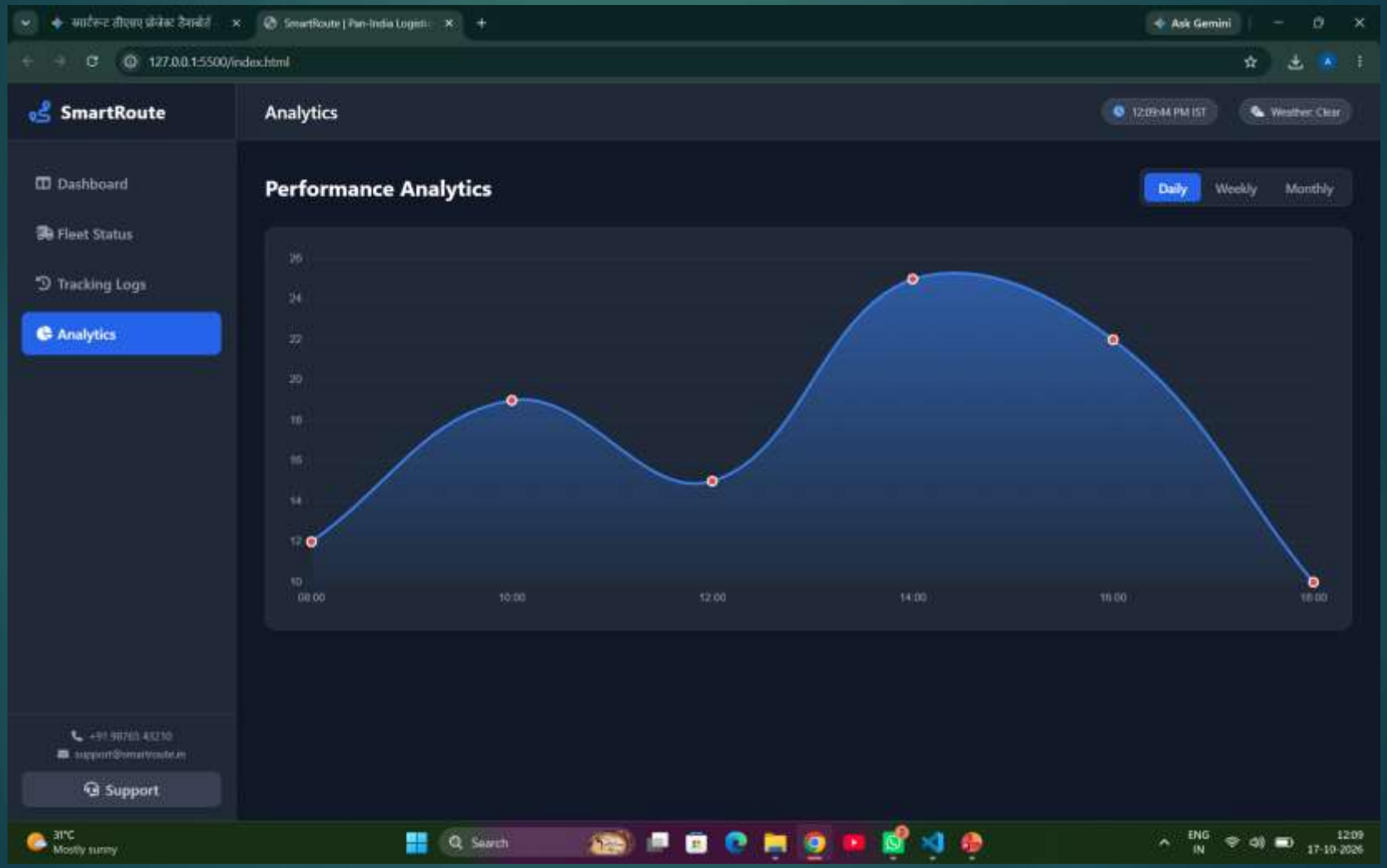
Dynamic Route Analytics & ETA



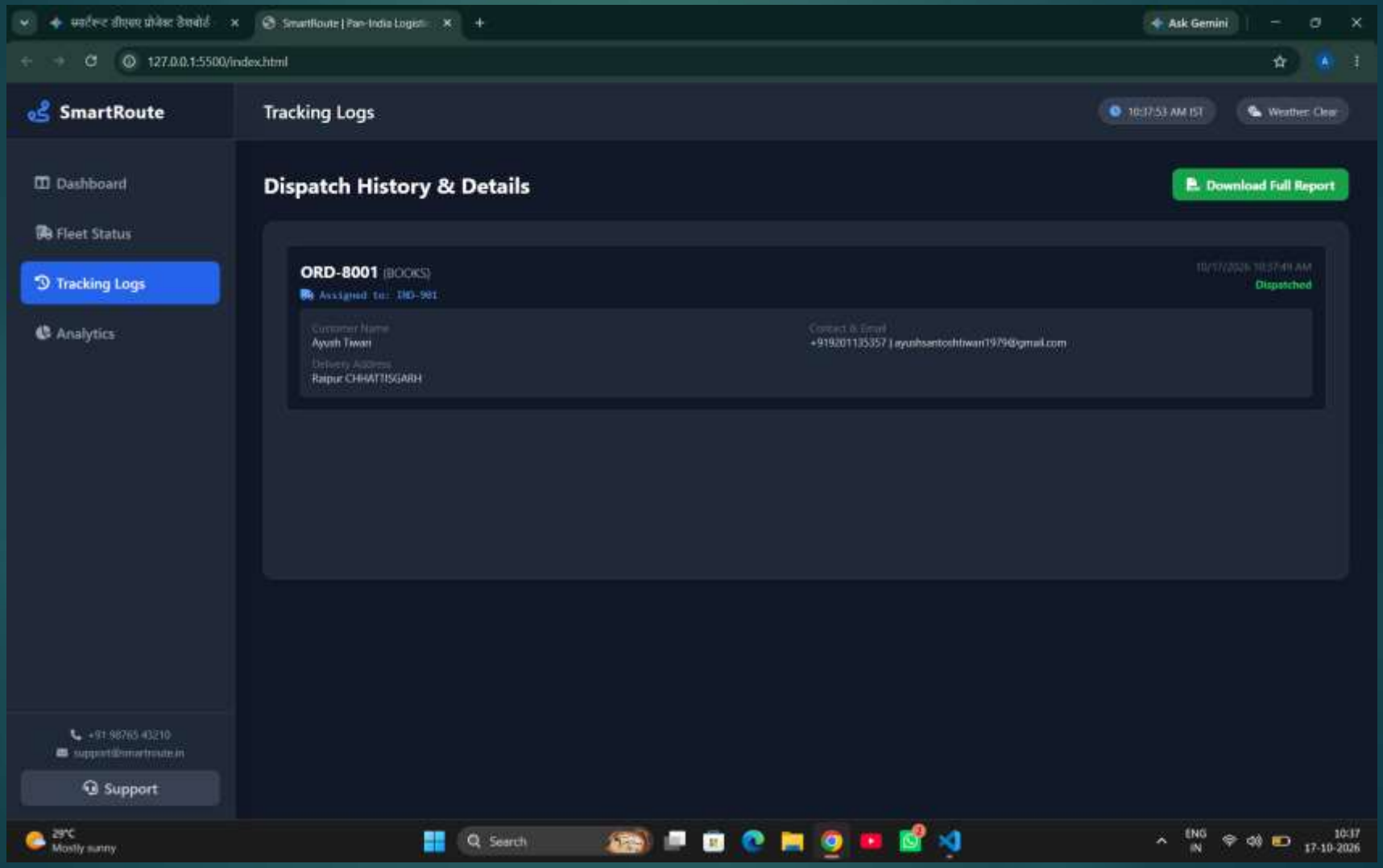
Advanced Features & Visual Simulations

- **Real-World Traffic & Weather Toggles**
  - Simulates roadblocks and storms by dynamically increasing graph edge weights, altering the ETA, speed, and cost of routes in real-time. Displays visual glow zones on the map.
- **Batch Dispatching & Fleet Engine**
  - Auto-assigns the top priority orders to available fleet vehicles. Features automated fuel tracking and status updates (Idle vs. In Transit).
- **Comprehensive Export**
  - One-click CSV export functionality that downloads highly detailed dispatch history reports directly to the user's local machine.

Performance Analytics (Smooth Line Graph)



Dispatch History & Detailed Logs



Technology Stack

- **Frontend Interface & Styling**
  - HTML5, Tailwind CSS, and Font Awesome used to create a sleek, responsive, dark-themed command center UI.
- **Mapping & Data Visualization**
  - Leaflet.js configured with OpenStreetMap tiles for zero-API-cost mapping, and Chart.js for rendering smooth gradient performance analytics.
- **Core Algorithms & Backend Logic**
  - 100% Vanilla JavaScript handling custom Graph construction, Min-Heap arrays, state management, and real-time clock mechanics.